QPCODE: **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) DEGREE EXAMINATION
THIRD SEMESTER

(2024 ADMISSION ONWARDS)

Discipline Specific Core Course (DSC B)

MG3DSCCHE202: INORGANIC AND ORGANIC CHEMISTRY

Duration: 1.5 Hr **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A – Multiple Choice Questions **(1×10 = 10 marks)**

(Answer all questions. Each question carries 1 mark).

1. Which of the following is a radioactive isotope? [CO 1] [U]

a) C-12 b) C-14

c) O-16 d) H-1

2. Which unit is used to measure radioactivity? [CO 1] [K]

a) Watt b) Becquerel

c) Newton d) Pascal

3. Give an example of iron-Sulphur protein [CO 2] [U]

a) Cytochrome b) Ferredoxin

c) Cobalamin d) Haemoglobin

4. Which is the oxygen-carrying protein in blood? [CO 2] [U]

a) Insulin b) Myosin

c) Hemoglobin d) Actin

5. What is the hybridization of the ring atom in an aromatic compound? [CO 3] [U]

a) sp b) sp^2

c) sp^3 d) None of the above

6. Which of the following is an example of polycyclic aromatic compound? [CO 3] [U]

a) Furan b) Pyridine

c) Pyrrole d) Anthracene

7. What is the role of BHA in food? [CO 4] [An]
a) Sweetener b) Preservative
c) Antioxidant d) Color enhancer
8. Which of the following is a non-permitted food color? [CO 4] [k]
a) Sunset Yellow b) Erythrosine
c) Metanil Yellow d) Indigo Carmine
9. Which of the following compounds shows optical isomerism? [CO 5] [U]
a) Ethane b) Butane
c) Lactic acid d) Cyclohexane
10. What is the shape of cyclohexane in its most stable form? [CO 5] [K]
a) Chair b) Boat
c) Planar d) Linear

Part B – Short Answer Questions (4×3 = 12 marks)

(Answer any four questions. Each question carries 3 marks.)

11. Write a note on natural and induced radioactivity. [CO 1] [U]
12. Describe the use of isotopes in radiotherapy. [CO 1] [U]
13. What is the role of myoglobin in the human body? [CO 2] [K]
14. Explain the effect of junk food on human health. [CO 4] [An]
15. Discuss the structure and stability of benzene. [CO 3] [U]
16. Draw and Explain the important conformation of n-butane [CO 5] [U]

Part C – Short Essay Questions (2×7 = 14 marks)

(Answer any two questions. Each question carries 7 marks.)

17. Discuss the structure, function, and importance of hemoglobin and chlorophyll. [CO 2] [U]
18. Explain the mechanism of aromatic electrophilic substitution reactions with examples. [CO 3] [K]
19. Describe the concept of optical isomerism with reference to tartaric acid. [CO 5] [K]

Part D – Essay Type Questions (1×14 = 14 marks)

(Answer any one question. Each question carries 14 marks.)

20. Explain the different types of stereo isomerism possible. Discuss about them with one example each [CO 5] [U]
21. Explain in detail nuclear fission, fusion, and applications of nuclear chemistry in medicine and dating techniques. [CO 1] [U]